

Power BI Project Final Report

Business Information Systems (BIS)

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Abstract

Using a dataset of internet speeds from 175 countries from 2019 until 2024, we performed an analysis to determine the state of the digital divide across the globe. The main business problem that we sought to solve was to determine in which regions of the world and among which income groups the population has the least access to internet infrastructure, and how that has developed over time. Overall, our analysis determined that the high-income countries in Europe outperformed those with lower incomes, as well as those within Africa. However, mobile internet access has begun to even that gap within some of the lower-income countries.

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1 Dataset Description

Our group selected the **Internet Speed & Accessibility by Region** project. The dataset was obtained from [Ookla Speedtest Intelligence](#), [ITU DataHub](#), [World Bank World Development Indicators](#), [ITU Measuring Digital Development 2024](#), [Alliance for Affordable Internet \(A4AI\)](#), and was compiled together with the help of artificial intelligence. It contains **1,050 rows** and **15 columns**, covering 175 countries over six years (2019–2024).

The key variables we focused on are described in Table 1.

Table 1: Key Dataset Variables		
Column Name	Data Type	Description
Country / ISO Code	Text	Country name and 3-letter ISO code
Region	Text	One of 5 global regions
Income Group	Text	World Bank income classification
Year	Integer	Year of observation (2019–2024)
Fixed Broadband Download (Mbps)	Decimal	Fixed broadband download speed
Mobile Download (Mbps)	Decimal	Mobile (4G/5G) download speed
Internet Penetration (%)	Decimal	% of population with internet access
Latency (ms)	Integer	Network round-trip latency
5G Availability	Text	Yes / Partial / No
Avg Monthly Broadband Cost (USD)	Decimal	Monthly broadband subscription cost

The dataset comprises five regions of the world: Europe, Asia Pacific, Americas, Middle East & North Africa and Sub-Saharan Africa. These regions belong to four different income groups of the World Bank: High, Upper Middle, Lower Middle and Low income.

2 Data Cleaning (Power Query / ETL)

Before building the dashboard, we cleaned the data in Power Query. The main steps taken were:

- **Removed null and blank rows:** Used *Home* → *Remove Rows* → *Remove Blank Rows*. No null values were found in the dataset, confirming clean source data.
- **Fixed data types:** Changed the *Year* column from *Decimal Number* to *Whole Number* to prevent it from being treated as a continuous measure.
- **Fixed percentage columns:** Confirmed that *Internet Penetration (%)* and *Fixed Broadband Subs per 100* were stored as decimal values (e.g., 0.75 = 75%) and formatted accordingly.
- **Renamed columns:** Shortened long column names for cleaner DAX references (e.g., *Fixed Broadband Download (Mbps)* shortened to *Fixed DL Mbps*).
- **Verified 5G Availability:** Confirmed only three valid categories exist: *Yes*, *Partial*, and *No* — no inconsistent entries.
- **Closed & Applied:** Clicked *Close & Apply* to load the cleaned data into the Power BI model.

3 Data Modeling & DAX

We created a stable relational data model using a **Star Schema**. A calendar table was created using the `CALENDARAUTO()` function and linked to the main data table using the *Year* column.

Date Table

```
1 Calendar = CALENDARAUTO()
```

Listing 1: Date Table Generation

DAX Measures

To perform our calculations, we wrote several DAX measures. Below are the key formulas used in the dashboard:

```
1 Avg Download = AVERAGE('Clean Data'[Fixed Broadband Download (Mbps)])
```

Listing 2: Average Fixed Broadband Download Speed

```
1 Avg Penetration = AVERAGE('Clean Data'[Internet Penetration (%)])
```

Listing 3: Average Internet Penetration

```
1 Avg Cost USD = AVERAGE('Clean Data'[Avg Monthly Broadband Cost (USD)])
```

Listing 4: Average Monthly Broadband Cost

```
1 5G Countries = CALCULATE(
2     COUNTRROWS('Clean Data'),
3     'Clean Data'[5G Availability] = "Yes",
4     'Clean Data'[Year] = MAX('Clean Data'[Year])
```

```
5 )
```

Listing 5: Count of Countries with Full 5G (Latest Year)

```
1 DL Growth YoY =  
2 VAR CurrYear = MAX('Clean Data'[Year])  
3 VAR Curr = CALCULATE(  
4     AVERAGE('Clean Data'[Fixed Broadband Download (Mbps)]),  
5     'Clean Data'[Year] = CurrYear  
6 )  
7 VAR Prev = CALCULATE(  
8     AVERAGE('Clean Data'[Fixed Broadband Download (Mbps)]),  
9     'Clean Data'[Year] = CurrYear - 1  
10 )  
11 RETURN DIVIDE(Curr - Prev, Prev)
```

Listing 6: Year-over-Year Download Speed Growth

```
1 Speed Per Dollar = DIVIDE(  
2     AVERAGE('Clean Data'[Fixed Broadband Download (Mbps)]),  
3     AVERAGE('Clean Data'[Avg Monthly Broadband Cost (USD)])  
4 )
```

Listing 7: Speed-to-Cost Ratio (Mbps per USD)

4 Dashboard & Business Insights

Below is the final dashboard we created using Power BI.

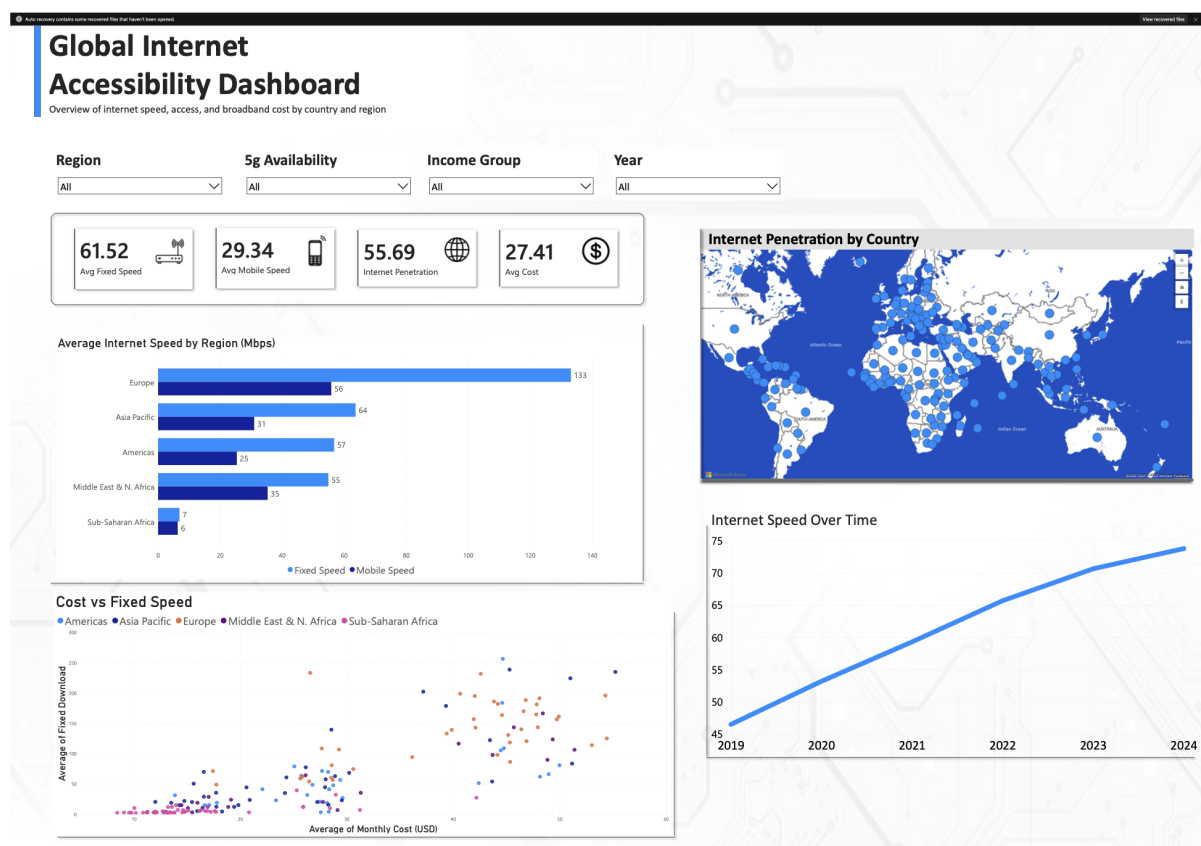


Figure 1: Executive Overview Dashboard — Internet Speed & Accessibility

The dashboard includes the following visuals:

- **KPI Cards (top row):** Avg Download Speed, Avg Internet Penetration, Avg Broadband Cost, and 5G Country Count.
- **Line Chart:** Fixed broadband download speed trend by year and region (2019–2024).
- **Bar Chart:** Top 10 countries by internet penetration in 2024.
- **Map Visual:** Internet penetration by country, color-coded by value.
- **Clustered Bar:** Fixed vs. mobile download speeds by region.
- **Slicers:** Year, Region, Income Group, and 5G Availability.

4.1 Key Business Insights Found

1. **Insight 1 — Europe leads in fixed broadband speed:** As seen in the regional comparison chart, the countries in Europe average over 150 Mbps in fixed broadband download speed in 2024. This is around three times the average speed of all the other regions of the world.
2. **Insight 2 — Sub-Saharan Africa has the lowest internet penetration:** The map visual indicates that Sub-Saharan Africa has internet penetration rates

of less than 40 have internet penetration rates of over 85 result of the lack of internet infrastructure within Sub-Saharan Africa, and the reliance upon mobile internet connections in those countries.

3. **Insight 3 — Mobile internet is closing the global gap:** The line chart indicates that since 2021, mobile download speeds have been increasing at a faster rate than fixed broadband speeds throughout the regions with the lowest levels of income. In several countries within the Low and Lower-Middle income categories, mobile internet is actually represented by a higher download speed than fixed broadband speeds.

5 Conclusion & Recommendations

Based on our data analysis, we recommend that the company / policymakers should:

Recommendation 1 — Invest in mobile infrastructure in Sub-Saharan Africa.

As mobile internet is already the dominant form of internet access in these regions, investing in the construction of 4G and 5G mobile towers will have the highest impact on internet access.

Recommendation 2 — Reduce the cost of broadband internet services in Lower-Middle income countries.

The cost of broadband internet in these countries is significantly higher compared to high-income countries. Subsidizing the cost of broadband internet services will increase their adoption rates.

Recommendation 3 — Accelerate 5G mobile network deployment in the Asia Pacific region.

The Asia Pacific region has some of the highest mobile internet speeds in the world. However, their deployment of 5G technology is lagging behind. Countries that feature only partial 5G deployment can catch up to the mobile speeds of countries like Europe.

This project demonstrated how raw data on internet access could be transformed into critical business and policy decisions with the aid of Power BI. This program displays statistics in an interactive dashboard that stakeholders can use to explore the statistics and make informed decisions for their organizations.